

ORIGINAL RESEARCH

Wavelet CNN-LSTM time series forecasting of electricity power generation considering biomass thermal systems

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Abstract

The use of biomass as a renewable energy source for electricity generation has gained attention due to its sustainability and environmental benefits. However, the intermittent electricity demand poses challenges for optimizing electricity generation in thermal systems. Time series forecasting techniques are crucial in addressing these challenges by providing accurate predictions of biomass availability and electricity generation. Here, wavelet transform is applied for denoising, convolutional neural networks (CNN) are used to extract features of the time series, and long short-term memory (LSTM) is applied to perform the predictions. The result of the mean absolute percentage error equal to 0.0148 shows that the wavelet CNN-LSTM is a promising machine-learning methodology for electricity generation forecasting. Additionally, this paper discusses the importance of model evaluation techniques and validation strategies to assess the performance of forecasting models in real-world applications. The major contribution of this paper is related to improving forecasting using a hybrid method that outperforms other models based on deep learning. Finally, future research directions and potential advancements in time series forecasting for biomass thermal systems are outlined to foster continued innovation in sustainable energy generation.

1 | INTRODUCTION

Electricity power generation through biomass thermal systems harnesses the renewable energy stored in organic materials. These systems utilize combustion or gasification processes to convert biomass into energy, which is then used to create steam to drive turbines and generate electricity power [1]. Biomass thermal systems offer a sustainable alternative to fossil fuels. Furthermore, they provide opportunities for waste management and agricultural diversification, turning organic waste into valuable energy resources [2]. The potential to predict the availability of energy in these systems can be valuable information for decision making, which is why time series forecasting is a promising solution that needs to be explored.

Biomass generation represents a sustainable alternative when hydroelectric generation systems do not meet demand and need

to be supplemented by other sources of generation. Considering that solar and wind energy depends on environmental conditions, one alternative is to use biomass as a generation source [3]. However, the cost of generation using thermal energy can be higher than other sources of generation, such as hydroelectric [4]. Based on this challenge, the model proposed here aims to forecast the generation of thermal energy using biomass to aid decision making regarding the management of this resource.

Time series forecasting of electricity generation in thermal systems powered by biomass involves predicting future electricity production based on historical data patterns [5]. These forecasts are crucial for optimizing energy production, managing resources efficiently, and planning for future demand [6]. By analysing past trends in biomass consumption, weather conditions, plant efficiency, and other relevant factors, predictive models can be applied to anticipate future electricity generation levels. This enables stakeholders to decide on energy

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production, distribution, and investment strategies. Additionally, advancements in machine learning (ML) and data analytics have enhanced the accuracy and reliability of these forecasts [7], facilitating sustainable energy management practices and promoting the integration of renewable energy resources into the energy grid [8].

A challenge in applying these models is the selection of the appropriate architecture. There are models based on deep learning (DL), promising alternatives in this field [9]. Besides the models that require greater computational effort, shallow models have also been applied in this field of study [10]. Among the prediction approaches, hybrid models have been a promising alternative in the forecasting of time series. The extraction of characteristics through convolutional neural network (CNN) along with forecasting approaches such as long short-term memory (LSTM) result in models that overcome classical structures and are being increasingly explored [11].

CNNs excel at capturing spatial patterns and local features, while LSTMs specialize in modelling temporal dependencies. By integrating these two components, the CNN-LSTM architecture can effectively handle noisy or complex time series data, making it more robust to variations and fluctuations in the input data [12]. This robustness is particularly valuable in real-world applications where data may be incomplete or subject to noise [13]. Considering the large nomenclature in this field, the study employs abbreviations as outlined in Table 1.

Enhancing the model's predictive capabilities by exploring denoising techniques is a promising way to yield improvements in the prediction models. Refining data preprocessing methodologies to better handle the characteristics of biomass thermal systems data could lead to more accurate forecasts. In this field, the wavelet transform has stood out as a method that allows signal noise to be reduced without loss of characteristics [14].

A benefit of the CNN-LSTM model is its capability for automatic feature extraction. Unlike standard forecasting methods that often require feature engineering, CNNs can learn relevant features automatically from the input sequences [15]. This reduces the need for domain expertise and human intervention in the feature extraction process, making the model more scalable and adaptable to different types of time series data [16].

The wavelet transform can be incorporated into the model to reduce signal noise, improving its predictive capacity. Considering the advantages of these approaches, this paper proposes a wavelet CNN-LSTM architecture for forecasting electricity power generation, especially considering biomass thermal systems. The major contributions of this paper are:

- The first contribution is regarding the improvement of the decision making considering the information of the predictions of electricity power generation in biomass thermal systems.
- The second contribution is encouraging the use of renewable sources, with greater control over their generation capacity about the necessary demand, given the recorded historical time series.

TABLE 1 Abbreviations and acronyms used.

Name	Abbreviations
Adaptive moment estimation	Adam
Artificial intelligence	AI
Artificial neural network	ANN
Autoregressive integrated moving average	ARIMA
Bidirectional LSTM	BiLSTM
Brazilian National System Operator	ONS
Convolutional neural network	CNN
Deep learning	DL
Deep non-parametric time series forecaster	DeepNPTS
Empirical wavelet transform	EWTF
Gated recurrent unit	GRU
Hidden units	HU
Interquartile range	IQR
Learning rate	LR
Long short-term memory	LSTM
Machine learning	ML
Mean absolute error	MAE
Mean absolute percentage error	MAPE
Neural basis expansion analysis	NBEATS
Neural hierarchical interpolation for time series	NHITS
One dimensional	1D
Root mean square error	RMSE
Root mean square propagation	RMSprop
Stochastic gradient descent	SGD
Seasonal ARIMA	SARIMA
Sequence-to-sequence	Seq2Seq
Singular spectral analysis	SSA
Symmetric MAPE	SMAPE
Temporal convolutional networks	TCN
Variational mode decomposition	VMD

- The third contribution is the combination of the wavelet transform, the CNN, and the LSTM, by using both architectures combined, the hybrid approach proved to be better than other DL models.

The remainder of this work is organized as follows: In Section 2, related works regarding time series forecasting are presented. Some solutions applied to improving the supply and management of power quality are presented first, followed by works focusing on applications of the CNN and LSTM models. Section 3 presents the methodology of the proposed method. In Section 4, the results and discussion are presented. In Section 5, a conclusion is presented and some future works are suggested.

2 | RELATED WORKS

The improvement of power network quality and reliability through the application of artificial intelligence (AI) represents a transformative leap in modern grid management [17]. AI algorithms, such as ML and deep neural networks, analyse historical records to identify patterns and anomalies in the power network [18], by predicting equipment failures, optimizing distribution, and detecting potential disturbances [19]. AI-powered predictive maintenance minimizes downtime, reduces costs in the industry, and extends the lifespan of critical infrastructure components [20]. Monitoring and control of the power system by AI enables to mitigate the impact of outages and voltage fluctuations, ensuring consistent and high-quality power supply to consumers. The use of AI methods has been highlighted mainly concerning forecasting [21].

Forecasting of time series in power systems is promising for predicting future electricity demand and production patterns, allowing electric utilities to effectively manage their resources and maintain grid stability [22]. By analysing historical data on factors such as weather conditions, economic trends, and past energy consumption, time series models can predict future load demand. These forecasts are essential for optimizing production schedules, planning maintenance activities, and making informed decisions about infrastructure investments [23]. This approach is important in integrating renewable energy sources, such as solar and wind power, into the grid by predicting their variable output. Advanced techniques, such as ML algorithms and neural networks, are increasingly being used to increase the accuracy and robustness of these forecasts, ensuring the reliable and efficient operation of energy systems [24].

Jin et al. [25] introduced an ensemble method for forecasting wind energy time series, utilizing a 15-min time window and forecast horizons of 4, 8, and 16 steps ahead. Their strategy achieved a root mean square error (RMSE) of 0.7771 MW for the 4-step-ahead forecast. When compared to several other models, their proposed method demonstrated superior performance, yielding lower errors than both the persistence method and Gaussian process regression. Suárez-Cetrulo et al. [26] evaluated a diverse set of machine-learning algorithms across eight wind farms to identify those with the lowest forecasting errors. The study tested 15 algorithms, including basic regression models, artificial neural networks (ANNs), and advanced ensemble techniques. The authors found that extreme gradient boosting and gradient boosting algorithms consistently outperformed the others in all the wind farms examined.

Stefenon et al. [27] proposed an approach to enhance the accuracy of insulator contamination forecasting by developing a hybrid wavelet stacking ensemble model. Insulator contamination in power transmission systems can cause critical failures and disrupt the electricity supply, highlighting the need for reliable forecasting methods. Their approach involves applying wavelet transformation to extract relevant features from historical contamination data, which are then integrated into a stacking ensemble framework. The hybrid wavelet stacking ensemble model presents a promising solution for addressing insulator contamination issues in modern energy infrastructure.

In the work of Safarian et al. [28], an ANN with a thermodynamic balance approach for downdraft biomass gasification with power generation units is presented. The model inputs encompass elemental analysis compositions, proximate analysis compositions, along operational parameters. Training data comprising 1032 simulated entries derived from 86 distinct biomass types across various operating conditions were utilized to train the ANN. The developed ANN demonstrates strong agreement with simulated data, achieving an absolute fraction of variance exceeding 0.999 for product power prediction. The proposed integrated ANN-based model holds significant potential as a practical tool for optimizing and controlling the downdraft biomass gasification with power generation units. Several papers present interesting proposals for the application of AI models, but having representative data can be challenging [29]. A database that can be highlighted is presented in reference [30], the dataset comprises experiments conducted on a biomass downdraft gasifier linked with a power production unit.

Enhancing forecasting through the synergistic fusion of ML, singular spectral analysis (SSA), and variational mode decomposition (VMD) represents a significant advancement in renewable energy prediction. By amalgamating these techniques, a comprehensive approach emerges, capable of capturing complex spatiotemporal patterns inherent in wind data [31]. ML algorithms, including DL models, can discern intricate relationships within the data, facilitating accurate predictions [32]. According to Moreno et al. [31], SSA aids in decomposing the time series into interpretable components, while VMD enhances the decomposition process by providing adaptability to non-stationary signals. The integration of these methodologies enables a holistic understanding of wind behaviour, resulting in refined forecasts crucial for optimizing wind energy generation and grid management.

A novel approach to forecasting electricity spot prices in Italy by integrating the Prophet and seasonal trend decomposition is proposed in reference [33]. By combining the strengths of both methods, the proposed approach aims to capture the complex patterns and fluctuations inherent in electricity spot prices, thereby enhancing forecast accuracy. Klaar et al. [34] applied ensemble learning approaches with seasonal decomposition to enhance energy price forecasting. In these studies, decomposition methods helped to reduce abrupt variations in the time series, making them a promising solution to be applied in this context.

In time series analysis, the decomposition of high frequencies refers to the process of breaking down the data into its constituent components, particularly focusing on the rapid fluctuations or variations occurring at higher frequencies within the time series [35]. These high-frequency components often represent short-term fluctuations or noise within the data that can obscure underlying patterns or trends [36]. By time series denoising, high frequencies can be eliminated and numerous techniques can be used like Fourier analysis, Savitzky–Golay filter [37], wavelet decomposition [14], empirical wavelet transform (EWT) [38], variational mode decomposition [39], kernel adaptive filter [40], or Hodrick–Prescott filter [41]. The decomposition and denoising aid in better understanding the

TABLE 2 Time series forecasting of electricity power generation.

Ref.	Method	Application
[43]	LSTM	Short-term wind power prediction
[44]	LSTM	Daily average load demand forecasting
[45]	Seq2Seq-LSTM with attention	Electricity load forecasting in Brazil
[46]	CNN-LSTM	Predicting short-term photovoltaic production
[47]	CNN-LSTM	Optimal demand response selection
[48]	CNN-LSTM	Solar power forecasting
[49]	CNN-LSTM	Short-term load forecasting
[50]	CNN-LSTM	Short-term household load forecasting
[51]	CNN-LSTM auto-encoder	Energy consumption forecasting
[52]	CNN-LSTM with attention	Load forecasting for combined heat and power use
[53]	LSTM with attention	Electricity price forecasting
[54]	Convolutional LSTM	Predicting household electric consumption
[55]	CNN-LSTM-BiLSTM with attention	Short-term electric load forecasting
[56]	Wavelet and multi-layer LSTM	Electricity consumption forecasting
[57]	BiLSTM with attention	Predicting energy consumption
[58]	LSTM and SARIMA	Electricity consumption forecasting
[59]	ARIMA, LSTM, GRU, ARIMA-LSTM, and ARIMA-GRU	Electrical energy consumption prediction
[60]	CNN-LSTM	Solar power prediction
[61]	LSTM-CNN with attention	Short-term load forecasting
[62]	EWT and TCN	Short-term electric load forecasting
[63]	TCN with attentional	Short-term load prediction
[64]	Extreme gradient boosting improved with Jaya optimizer	Forecasting energy consumption in residential buildings

underlying structure and dynamics of the time series, facilitating more accurate forecasting and decision making in various domains [42].

Currently, LSTM models combined with decomposition techniques are gaining ground in energy forecasting and have been applied by various authors. Some examples of these applications are shown in Table 2. Besides the standard LSTM models, some researchers are evaluating gated recurrent unit (GRU), autoregressive integrated moving average (ARIMA), bidirectional LSTM (BiLSTM), temporal convolutional networks (TCN), sequence-to-sequence (Seq2Seq) time series data, among other variations that are showing promise in this field. Even the classic seasonal autoregressive integrated moving average (SARIMA) model can be applied in this regard to enhance the predictions.

Crop yield forecast using CNN-LSTM evaluated by Nejad et al. [65] showed some advantages as decreased network complexity however your limitations occurred because of the software and hardware limitations to processing the remote sensing images with high-dimensional data information consequently providing spatial-spectral configuration and verifying higher resolution in the pictures and tuning hyperparameters to reduce errors in the prediction.

Regarding biomass estimation, Kijo-Kleckowska et al. [66] investigated the impact of various base fluids on the convective heat transfer properties within solar collectors utilizing

nanofluids. The researchers analysed how different base fluids, including water, ethylene glycol, and oil, influenced the thermal conductivity and overall efficiency of solar energy systems when combined with nanoparticles. The study's findings highlighted the significant role that the selection of base fluids played in optimizing solar collector performance, suggesting that specific base fluids could be tailored to enhance energy capture and transfer.

Lee, Choi, and Hwang [67] presented a study of ML applied in bio-oil research. According to them, regression methods are the most commonly used for building predictive models, with the top five ML techniques being random forest, ANNs, gradient boosting, support vector regression, and linear regression. Current research faces challenges such as limited data availability, feature extraction based on domain knowledge, and restricted use of ML.

In reference [68], the prediction using ML was focused on the evaluation of the most important parameters setup for power generation. Considering several evaluations they were able to achieve a mean square error of 1497, being a promising result using ML. Samadi, Ghobadian, and Nosrati [69] evaluated the biomass energy from agricultural residues, and based on a stoichiometric equilibrium model achieved an RMSE from 0.571 to 1.10. Considering these results, the authors have shown how high can be the power generation capacity using agricultural residues.

Besides the standard LSTM that has been used by several authors, many works are applying the CNN-LSTM for time series forecasting, especially focused on energy. The attention mechanism is used by some authors with promising results as well. Considering the stability and the promising results of CNN-LSTM, this structure is combined with the wavelet transform and used here. The complete methodology of the wavelet CNN-LSTM is presented in the next section.

3 | METHODOLOGY

Time series forecasting plays an important role in decision-making processes across various industries [70]. Traditional methods may struggle to capture the intricate patterns present in real-world time series data [71]. With the advancements in DL, more complex architectures have been developed to handle the intricacies of time series data, such as the LSTMs [72]. The predictions in the LSTM network can be given according:

$$\hat{y}_{t+1} = f_{\theta}(x_t, b_t, c_t), \quad (1)$$

where $\hat{y}_{t+1}$ represents the forecasted value for the next time step $t + 1$, f_{θ} denotes the LSTM function with parameter θ , which takes input features x_t at time t , hidden state b_t at time t , and cell state c_t at time t to predict the next value. LSTMs are equipped with memory cells and gating mechanisms that allow them to selectively remember or forget data over long sequences. This capability makes LSTMs particularly suitable for time series forecasting tasks [73], where understanding long-range dependencies is crucial [74].

CNNs are powerful at capturing spatial patterns in data. Regarding time series forecasting, one-dimensional (1D) CNNs are employed to automatically learn relevant features from sequential data. These features may represent patterns or motifs within the time series, such as local trends or repeating structures [75]. The CNN layers operate by sliding a set of filters (kernels) over the input sequence, extracting features through convolutions, and applying non-linear activation functions [76].

The CNN layers serve as feature extractors, capturing relevant patterns from the input time series data. These extracted features are then fed into the LSTM layers, which model the temporal dynamics and dependencies to make accurate predictions [77]. By combining CNNs and LSTMs, the model can learn hierarchical representations of the input data, capturing both local and global patterns. This architecture can be represented by:

$$\begin{aligned} &\text{Input : Time Series Data} \\ &\text{1D Convolutional Layers : Extract spatial features} \\ &\text{LSTM Layers : Model temporal dependencies} \\ &\text{Output : Forecasted time series.} \end{aligned} \quad (2)$$

The LSTM layers in the CNN-LSTM architecture receive the feature representations extracted by the CNN layers and fur-

ther model the sequential nature of the data, learning to predict future values based on the learned patterns and dependencies [78]. By incorporating LSTM layers, the model can capture complex relationships and dependencies that span across multiple time steps. This ability to capture long-range dependencies is crucial for accurate forecasting, especially in scenarios where historical information significantly impacts future predictions [79].

In this architecture, for the LSTM, b_t and c_t denote the output and state of the cell, respectively. Considering the input port i for controlling the cell's status update level, the forget gate of the state reconfiguration control f , the candidate cell for adding information g , and the output gate o that controls the state level of the cell added to the hidden state, the next equations describe the components at time step t :

$$\begin{aligned} i_t &= \sigma_g(W_i x_t + R_i b_{t-1} + b_i), \\ f_t &= \sigma_g(W_f x_t + R_f b_{t-1} + b_f), \\ g_t &= \sigma_c(W_g x_t + R_g b_{t-1} + b_g), \\ o_t &= \sigma_g(W_o x_t + R_o b_{t-1} + b_o), \end{aligned} \quad (3)$$

where σ_g and σ_c are the gate and the cell activation functions. The learnable parameters of an LSTM layer include the input weights W , the recurrent weights R , and the bias b . The W , R , and b are, respectively, the concatenations of the input and recurrent weights, and the biases of each component, as follows:

$$W = \begin{bmatrix} W_i \\ W_f \\ W_g \\ W_o \end{bmatrix}, \quad R = \begin{bmatrix} R_i \\ R_f \\ R_g \\ R_o \end{bmatrix}, \quad b = \begin{bmatrix} b_i \\ b_f \\ b_g \\ b_o \end{bmatrix}. \quad (4)$$

To forecast a sequence, the training sequences' responses are advanced by a one time step. Consequently, at every point in the input sequence, the LSTM model grasps the task of predicting the subsequent time step's value [80]. Here, tree optimizers were used, the stochastic gradient descent (SGD), adaptive moment estimation (Adam), and the root mean square propagation (RMSprop). The SGD is given by the following equation:

$$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t),$$

where θ_t represents the parameters at time t , η denotes the learning rate, and $\nabla J(\theta_t)$ is the the loss function J with respect to the parameters θ and t . The Adam optimizer can be calculated as:

$$\begin{aligned} m_{t+1} &= \beta_1 m_t + (1 - \beta_1) \nabla J(\theta_t), \\ v_{t+1} &= \beta_2 v_t + (1 - \beta_2) (\nabla J(\theta_t))^2, \\ \hat{m}_{t+1} &= \frac{m_{t+1}}{1 - \beta_1^{t+1}}, \\ \hat{v}_{t+1} &= \frac{v_{t+1}}{1 - \beta_2^{t+1}}, \end{aligned}$$

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_{t+1}}{\sqrt{\hat{v}_{t+1} + \epsilon}},$$

where β_1 and β_2 are exponential decay rates for moment estimates, m_t and v_t are the first and second moment estimates, respectively, $\hat{m}_{t+1}$ and $\hat{v}_{t+1}$ are bias-corrected estimates, and ϵ is a small constant to prevent division by zero. The RMSprop is calculated by:

$$V_{t+1} = \beta V_t + (1 - \beta)(\nabla J(\theta_t))^2,$$

$$\theta_{t+1} = \theta_t - \eta \frac{\nabla J(\theta_t)}{\sqrt{V_{t+1} + \epsilon}},$$

where β is the decay rate and V_t is the exponentially decaying average of squared gradients.

By learning hierarchical representations, automatically extracting features, capturing long-term dependencies, and enhancing model robustness, CNN-LSTM provides a powerful framework for modelling complex temporal patterns [81]. Further research in CNN-LSTM and related architectures are expected to drive continued progress in time series forecasting applications across various domains [82]. Besides the CNN-LSTM, this paper proposes a filtering step. After the model has been adjusted, a filter based on the wavelet transform is applied for signal denoising. The architecture of the wavelet CNN-LSTM applied here is presented in Figure 1.

3.1 | Denoising

The wavelet transform is applied to denoise time series signals, reducing the signal's high frequencies. Its application is based on the decomposition and reconstruction of wavelet packets. In this procedure, the reconstructed signal has fewer non-linearities than the original signal, which is an advantageous characteristic to be applied in time series forecasting [27].

The wavelet packet technique is a branch of wavelet decomposition to extract signal features. Wavelet packets are waveforms indexed by position, frequency, and scale. The wavelet decomposition can be expressed as:

$$W_{\Psi, X}(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(t) \Psi * \left(\frac{t-b}{a} \right) dt, \quad a \neq 0, \quad (5)$$

where, $x(t)$ is the decomposed signal, a and b are, respectively, the scale and displacement parameters, and $\Psi(t)$ is the mother wavelet [83]. Considering a discretization, the mother wavelet $\Psi(n)$ and scaling function $\Phi(n)$ are given by:

$$\Psi(n) = \sum_{i=0}^{N-1} g(i) \Phi(2n-i), \quad (6)$$

$$\Phi(n) = \sum_{i=0}^{N-1} b(i) \Phi(2n-i), \quad (7)$$

where g is a high-pass filter and b is a low-pass filter [35]. The methodology for the decomposition and reconstruction of wavelet packets is presented in Figure 2.

3.2 | Benchmarking setup

A comparative analysis is carried out, in addition to the original CNN-LSTM model [84], the state-of-the-art deep non-parametric time series forecaster (DeepNPTS) [85], TCN [86], neural hierarchical interpolation for time series (NHITS) [87], and neural basis expansion analysis (NBEATS) [88] models are compared to the proposed wavelet CNN-LSTM model.

3.3 | Dataset

The Brazilian National System Operator (ONS) is responsible for coordinating and managing the operation of electricity generation and transmission facilities within the National Interconnected System. It also plans the operation of the country's isolated systems, all under the supervision and regulation of the National Electric Energy Agency [89]. The ONS comprises associate members and participating members, including generation, transmission, and distribution companies, free consumers, and energy importers and exporters.¹

Operation planning in the Brazilian electricity system is currently carried out in two stages, using DECOMP and the NEWAVE model, defined by ONS. DECOMP is an optimization computer model for planning the short-term energy operation of hydrothermal and wind systems. It can refine the operating policy constructed by the NEWAVE (medium-term model), considering more detailed aspects of hydroelectric power plants, thermoelectric power plants, and the transmission system [90].

Considering the uncertainty in inflows and risk aversion these models guarantee robust dispatch and, as a result, the most appropriate operating policy for the daily programming of the system's operation [91]. The DECOMP uses an observation time window of 3 months to make the hourly short-term forecast. Considering that the focus of this paper is short-term forecasting, this paper assumes the same period of analysis.

The database used is based on the verified generation of power plants, sets of power plants, and groups of small power plants on an hourly basis, made available by the ONS.² Especially for this case study, the data was filtered for biomass-based generation, with a total of 58 power plants connected to the national electricity power system. The power generation sum results in a time series that represents the total energy generated by biomass in Brazil, this time series is presented in Figure 3. For hourly evaluation of the variation in energy generated by biomass, 3 months were considered (containing a total of 2208 records, from 0 h of 9 January 2024, to 23 h of 9 May 2024).

¹ Information about the organization at: <https://dados.ons.org.br/organization/about/ons> (in Portuguese).

² For future analysis, the data can be found at: <https://dados.ons.org.br/dataset/geracao-usina-2>.

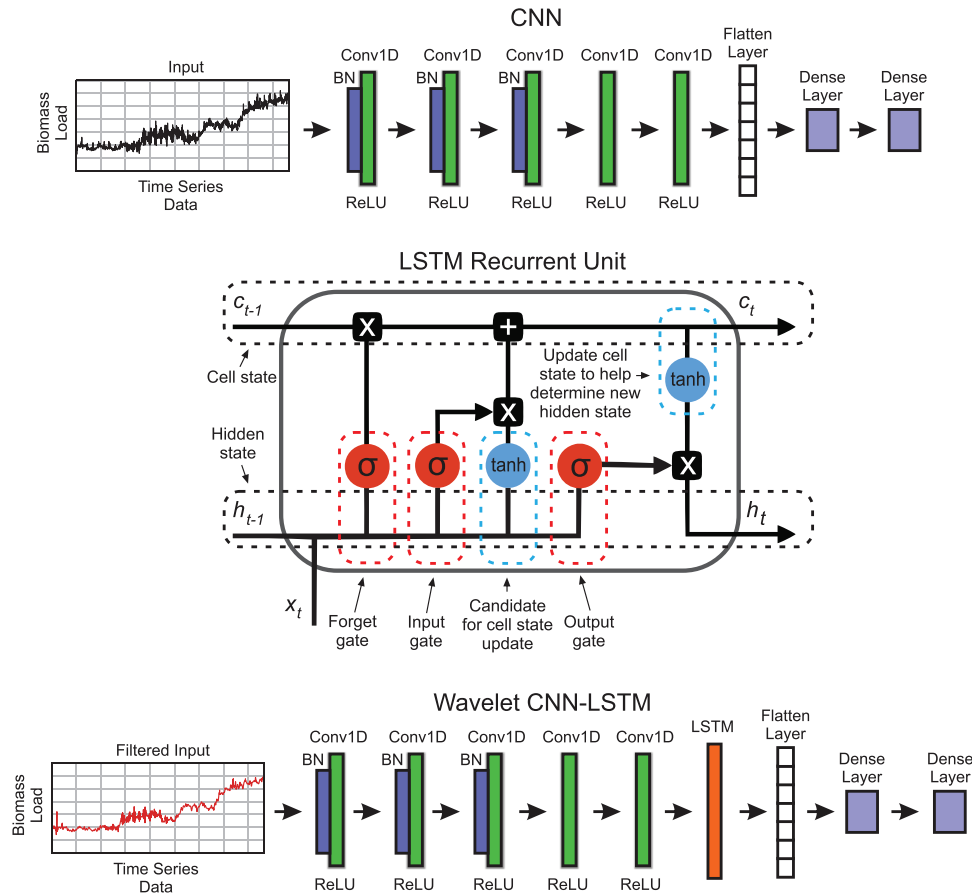


FIGURE 1 Architecture of the wavelet CNN-LSTM model.

As presented in Figure 3, biomass generation has great variability, which makes it more challenging to forecast, making it necessary to use input filters, as applied here. Evaluating other generation sources that have the role of improving system stability would be simpler as they suffer less variation over time, which is necessary for a reliable energy system. However, the goal of this paper is to propose a model that can deal with non-linearities.

From the total biomass thermal power plants installed in Brazil, five are located in the north, two in the northeast, one in the south, and 49 in the southeast. Considering that the southeast is the most industrialized region in the country, one of the main advantages of thermal power plants is the possibility of installing them close to major consumer centers. Since the national system is interconnected, the assessment of biomass generation presented here considers the sum of all 58 power plants.

4 | RESULTS

Here, the results of the wavelet CNN-LSTM are presented and commented on. The experiment setup for this evaluation is presented in Section 4.1. The results of the model assessment are presented in Section 4.2. In Section 4.3, a statistical assessment is evaluated considering 50 runs, where the mean, median,

mode, range, standard deviation, percentiles, and interquartile range (IQR) are presented. After the CNN-LSTM model has been adjusted, the filter is used as shown in Section 4.4, and finally benchmarking is carried out in Section 4.5.

4.1 | Experiment setup

The experiments were considering an NVIDIA T4 graphics processing unit with 16 GB of memory, the algorithm was written in Python, and the computations were done using the Google computing engine backend (colab.google). Here, the evaluation error performances were RMSE, mean absolute error (MAE), mean absolute percentage error (MAPE), and symmetric MAPE (SMAPE), and coefficient of determination (R^2) given by:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (8)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (9)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (10)$$

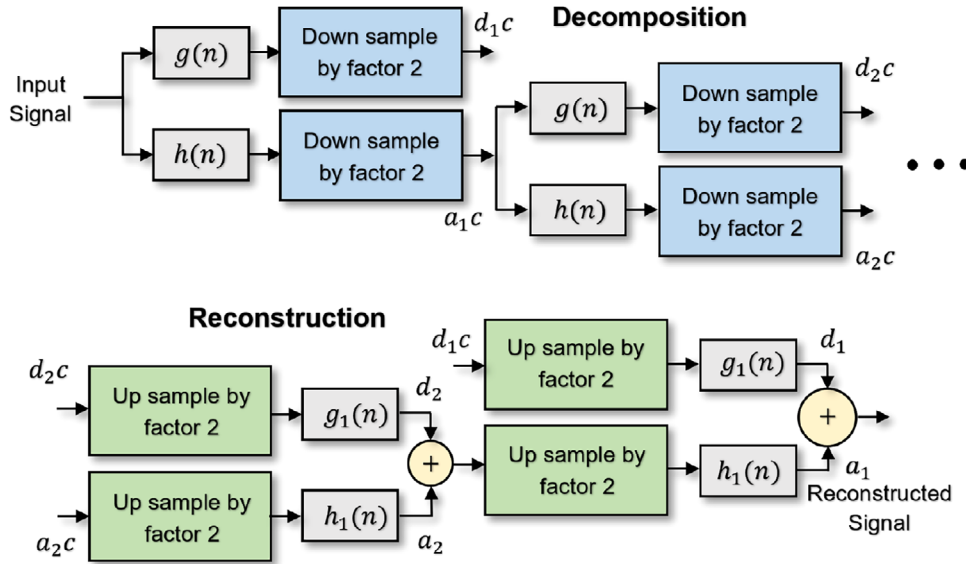


FIGURE 2 Decomposition and reconstruction of wavelet packets.

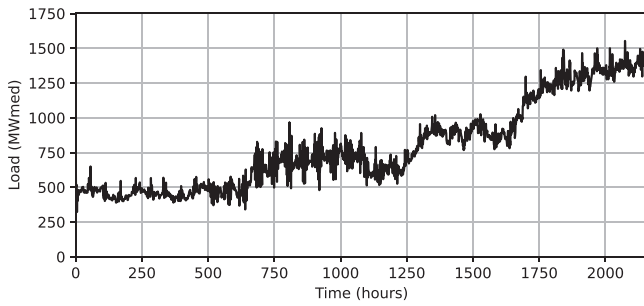


FIGURE 3 Biomass energy generation in Brazil over time.

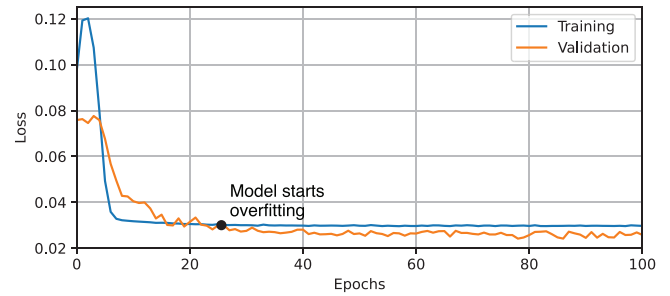


FIGURE 4 Losses during training.

$$\text{SMAPE} = \frac{100\%}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2}, \quad (11)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}, \quad (12)$$

where y_i represents the real value, $\hat{y}$ stands for the estimated value, and n signifies the total observations [92]. Considering that the model may start to overfit, the early stop is applied. When the validation loss does not improve compared to the training loss the model stops the training. For comparison, the maximum of 100 epochs is considered, and the early stop occurs when there are more than 20 epochs without improvement (patience). An example of the overfitting is presented in Figure 4.

4.2 | CNN-LSTM assessment

The evaluation presented here is focused on identifying the right parameters for the best commitment in the model. The

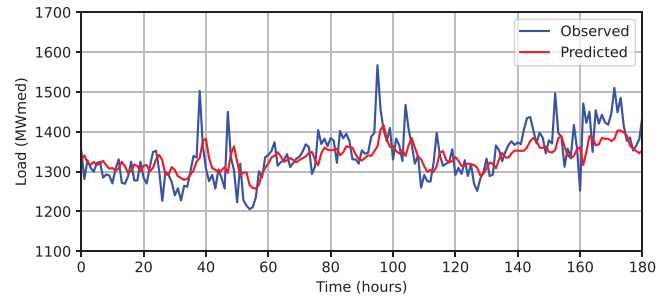


FIGURE 5 Comparative result of predicted by CNN-LSTM versus observed values.

analysis of the CNN-LSTM method starts with the percentage relationship between the data split for training and testing and the observed horizon for forecasting. An example of the results comparing the observed and predicted values from the CNN-LSTM is shown in Figure 5, considering 90% of the data for training and 10% of the data for testing.

The data split and the horizon can have a major influence on the results. Given that an evaluation, considering different horizons and different data splits, is presented in Table 3. The error

TABLE 3 Evaluation of the data split ratio for training and testing the model and forecast horizon.

Horizon	Ratio	RMSE	MAE	MAPE	SMAPE	R ²	Time (s)
5	70/30	2.98×10^2	2.54×10^2	1.83×10^{-1}	2.07×10^1	2.19×10^{-1}	1.44×10^2
	80/20	1.13×10^2	9.47×10^1	6.91×10^{-2}	7.25	1.47×10^{-1}	8.72×10^1
	90/10	5.69×10^1	4.20×10^1	3.05×10^{-2}	3.12	1.53×10^{-1}	4.49×10^2
10	70/30	3.16×10^2	2.70×10^2	1.95×10^{-1}	2.21×10^1	3.73×10^{-1}	1.47×10^2
	80/20	1.14×10^2	9.55×10^1	6.95×10^{-2}	7.29	2.75×10^{-1}	1.71×10^2
	90/10	7.88×10^1	6.40×10^1	4.63×10^{-2}	4.79	5.88×10^{-1}	3.68×10^2
15	70/30	2.90×10^2	2.48×10^2	1.79×10^{-1}	2.01×10^1	1.79×10^{-1}	2.93×10^2
	80/20	1.11×10^2	9.34×10^1	6.80×10^{-2}	7.12	3.11×10^{-1}	2.51×10^2
	90/10	5.26×10^1	3.83×10^1	2.79×10^{-2}	2.84	3.09×10^{-1}	1.93×10^2
20	70/30	2.63×10^2	2.27×10^2	1.66×10^{-1}	1.83×10^1	1.85×10^{-2}	8.15×10^1
	80/20	1.33×10^2	1.15×10^2	8.40×10^{-2}	8.88	1.00	1.81×10^2
	90/10	4.77×10^1	3.49×10^1	2.58×10^{-2}	2.59	4.30×10^{-1}	2.04×10^2
25	70/30	2.24×10^2	1.97×10^2	1.46×10^{-1}	1.57×10^1	2.71×10^{-1}	1.59×10^2
	80/20	1.49×10^2	1.34×10^2	9.82×10^{-2}	1.04×10^1	1.59	2.76×10^2
	90/10	5.16×10^1	3.89×10^1	2.92×10^{-2}	2.89	3.54×10^{-1}	2.05×10^2

evaluation metrics used here aim to verify the lowest possible prediction error, given the configuration parameters evaluated.

For this model, the use of different forecast horizons did not result in a significant change in performance, but the use of less training data had an impact on the model's performance. For all the horizons evaluated, the use of 90% of the data for training brought superior results with lower forecast error, considering that a forecast with 10% of the data is adequate for testing, this value was set for all the analyses.

The best horizon observed was 20 h, which is why this horizon was considered here to be the appropriate horizon for evaluating energy forecasting. Initially, for this experiment, the Adam optimizer was considered, with a learning rate equal to 1×10^{-3} , one layer, and four hidden units (HU). To obtain an optimized structure, these model configuration parameters were evaluated. Table 4 shows the results for varying the layers and HU.

The variation in the number of layers and HU resulted in a minor variation in the model results, and the initial configuration had a satisfactory outcome, showing that the model was initially well designed. In this comparative analysis, the best result was using two layers and one hidden unit for each of these layers, and this result was used as the standard architecture for the following analyses.

Using more layers and a deeper network strategy did not bring promising results, which is why this approach was not used. Considering the best data selection and model architecture, an analysis is presented in Table 5 to evaluate the optimizer and the learning rate (LR) used in the CNN-LSTM model, these being the last model parameters to be evaluated.

The use of an inappropriate LR considerably impairs the training of the model, and performance results are adversely affected when very high or low LR values are used. The best outcome in this evaluation was with the Adam optimizer and

an LR of 1×10^{-3} , which were the parameters initially used in this evaluation. This proves that the design of the architecture was well done and that the variation in the model's parameters resulted in a reduction in its performance.

The aim of the evaluation presented in Tables 3 to 5 is to define the best parameters for configuring the model so that it achieves its best performance. Specifically, the best data use configuration concerning the prediction horizon is evaluated in Table 3, while Table 4 shows the influence of changing the model's architecture (use of deeper layers and more hidden units), and finally, Table 5 evaluates the optimizer's settings.

The comparative results between the forecasted and observed values show that the model has greater difficulty in predicting abrupt variations, which is to be expected given that these variations are not expected, and the model has to use the historical average values for training. A possible way to improve these results would be to use filters that attenuate the observed values, thus disregarding abrupt variations. Here, the original signals were considered so that a model could be directly used in this evaluation.

4.3 | Statistical assessment

Observing that there was variability in the results, considering a random initialization of the of the neural network weights, a statistical analysis is presented here. To assess the variability of the neural network, 50 experiments with random initialization, to assess whether the model presents a significant variation in its error results if more experiments are evaluated. The result of the statistical analysis considering these experiments is shown in Table 6. The statistical assessment presented in Table 6 aims to verify whether the parameters defined according to the results of Tables 3 to 5 are adequate for the model to be stable to

TABLE 4 Evaluation of the number of layers and hidden units in the model architecture.

Layers	HU	RMSE	MAE	MAPE	SMAPE	R ²	Time (s)
1	1	5.34×10^1	3.97×10^1	2.89×10^{-2}	2.94	2.84×10^{-1}	3.70×10^2
	2	4.87×10^1	3.59×10^1	2.64×10^{-2}	2.66	4.05×10^{-1}	3.97×10^2
	3	5.10×10^1	3.77×10^1	2.76×10^{-2}	2.80	3.49×10^{-1}	1.99×10^2
	4	4.83×10^1	3.56×10^1	2.63×10^{-2}	2.64	4.16×10^{-1}	4.13×10^2
2	1	4.81×10^1	3.53×10^1	2.60×10^{-2}	2.62	4.21×10^{-1}	6.32×10^2
	2	4.82×10^1	3.56×10^1	2.62×10^{-2}	2.64	4.18×10^{-1}	6.77×10^2
	3	6.17×10^1	4.69×10^1	3.40×10^{-2}	3.48	4.44×10^{-2}	2.82×10^2
	4	5.94×10^1	4.48×10^1	3.25×10^{-2}	3.33	1.15×10^{-1}	5.86×10^2
3	1	8.31×10^1	6.75×10^1	4.89×10^{-2}	5.07	7.31×10^{-1}	3.28×10^2
	2	5.55×10^1	4.17×10^1	3.05×10^{-2}	3.09	2.28×10^{-1}	7.02×10^2
	3	7.03×10^1	5.50×10^1	3.98×10^{-2}	4.10	2.38×10^{-1}	3.67×10^2
	4	6.44×10^1	4.91×10^1	3.56×10^{-2}	3.65	3.84×10^{-2}	3.76×10^2
4	1	8.73×10^1	7.10×10^1	5.14×10^{-2}	5.34	9.10×10^{-1}	5.82×10^2
	2	6.56×10^1	5.01×10^1	3.63×10^{-2}	3.73	7.81×10^{-2}	8.41×10^2
	3	5.53×10^1	4.16×10^1	3.04×10^{-2}	3.08	2.34×10^{-1}	1.05×10^3
	4	5.95×10^1	4.49×10^1	3.26×10^{-2}	3.33	1.14×10^{-1}	8.94×10^2

TABLE 5 Optimizer evaluation and model learning rate.

Optimizer	LR	RMSE	MAE	MAPE	SMAPE	R ²	Time (s)
Adam	1×10^{-1}	1.30×10^2	1.14×10^2	8.33×10^{-2}	8.79	3.24	1.37×10^2
	1×10^{-2}	1.19×10^2	1.03×10^2	7.47×10^{-2}	7.85	2.56	6.06×10^2
	1×10^{-3}	4.74×10^1	3.45×10^1	2.56×10^{-2}	2.56	3.98×10^{-1}	4.35×10^2
	1×10^{-4}	6.09×10^1	4.65×10^1	3.37×10^{-2}	3.46	7.02×10^{-2}	6.17×10^2
	1×10^{-5}	6.53×10^2	6.50×10^2	4.84×10^{-1}	6.39×10^1	1.06×10^2	1.31×10^2
RMSprop	1×10^{-1}	7.72×10^1	6.41×10^1	4.88×10^{-2}	4.75	4.94×10^{-1}	5.17×10^2
	1×10^{-2}	7.96×10^1	6.18×10^1	4.48×10^{-2}	4.62	5.90×10^{-1}	1.16×10^2
	1×10^{-3}	7.81×10^1	6.09×10^1	4.40×10^{-2}	4.55	5.31×10^{-1}	3.28×10^2
	1×10^{-4}	9.91×10^1	8.46×10^1	6.15×10^{-2}	6.41	1.46	4.42×10^2
	1×10^{-5}	7.93×10^2	7.90×10^2	5.88×10^{-1}	8.34×10^1	1.56×10^2	5.30×10^2
SGD	1×10^{-1}	8.09×10^1	6.78×10^1	5.17×10^{-2}	5.01	6.40×10^{-1}	1.04×10^2
	1×10^{-2}	1.14×10^2	9.75×10^1	7.08×10^{-2}	7.43	2.27	1.82×10^2
	1×10^{-3}	1.36×10^2	1.20×10^2	8.79×10^{-2}	9.29	3.62	9.67×10^1
	1×10^{-4}	1.88×10^2	1.77×10^2	1.30×10^{-1}	1.40×10^1	7.88	4.53×10^2
	1×10^{-5}	7.52×10^2	7.50×10^2	5.58×10^{-1}	7.74×10^1	1.41×10^2	9.41×10^1

the variation in weights caused by the random initialization of the network.

For a clearer visualization of the variability of results in the 50 experiments, Figure 6 shows graphs representing the variability of the error for all experiments. As can be seen, the vast majority of results were close to the average, demonstrating that the model is reliable and is not affected by the random initialization of the neural network weights. Observing Figure 6, with an average MAPE of 0.0259 considering 50 runs, the proposed

method is very promising for the forecasting of electricity power generation considering biomass thermal systems.

Figure 6 clarifies that the model is stable since no significant outliers exist. This shows that the model has been properly configured and is not vulnerable to random initialization of the people in the network. Considering the model to be stable, the next subsection presents the results of using the wavelet transform filter as an input data step for the prediction model.

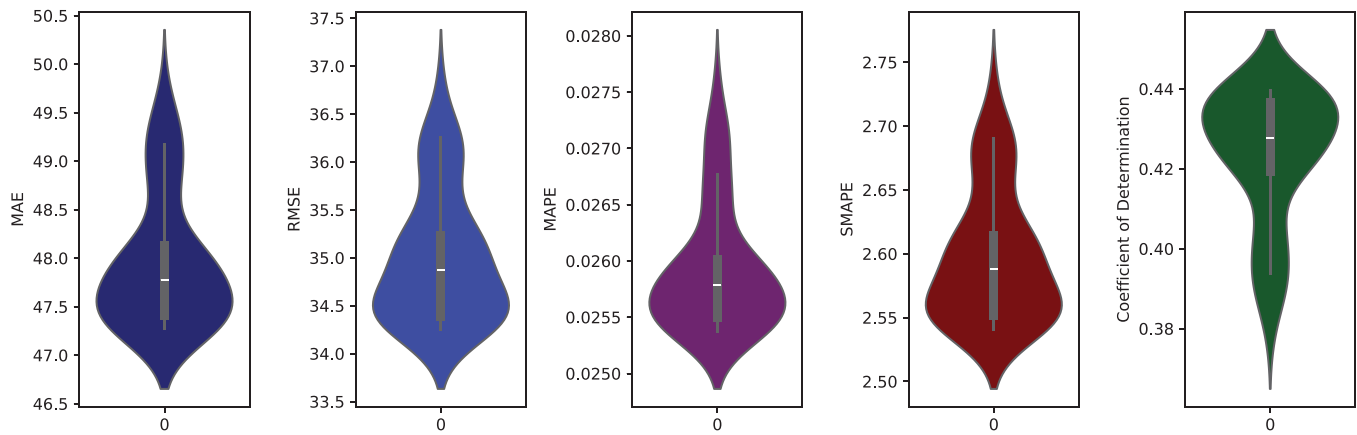


FIGURE 6 Violin plots of the evaluated errors considering 50 runs.

TABLE 6 Statistical assessment.

	RMSE	MAE	MAPE	SMAPE	R ²
Mean	4.80×10^1	3.50×10^1	2.59×10^{-2}	2.60	4.23×10^{-1}
Median	4.78×10^1	3.49×10^1	2.58×10^{-2}	2.59	4.28×10^{-1}
Mode	4.73×10^1	3.43×10^1	2.54×10^{-2}	2.54	3.80×10^{-1}
Range	2.45	2.50	2.17×10^{-3}	1.88×10^{-1}	5.95×10^{-2}
Std. Dev.	6.84×10^{-1}	6.82×10^{-1}	5.59×10^{-4}	5.10×10^{-2}	1.66×10^{-2}
25th %ile	4.74×10^1	3.44×10^1	2.55×10^{-2}	2.55	4.19×10^{-1}
50th %ile	4.78×10^1	3.49×10^1	2.58×10^{-2}	2.59	4.28×10^{-1}
75th %ile	4.81×10^1	3.52×10^1	2.60×10^{-2}	2.61	4.37×10^{-1}
IQR	7.17×10^{-1}	8.39×10^{-1}	5.08×10^{-4}	6.23×10^{-2}	1.72×10^{-2}

TABLE 7 Wavelet level analysis.

L	RMSE	MAE	MAPE	SMAPE	R ²
1	3.60×10^1	2.70×10^1	1.98×10^{-2}	2.00	6.49×10^{-1}
2	4.59×10^1	3.81×10^1	2.77×10^{-2}	2.82	3.49×10^{-1}
3	3.41×10^1	2.62×10^1	1.90×10^{-2}	1.93	6.40×10^{-1}
4	2.89×10^1	2.48×10^1	1.85×10^{-2}	1.84	6.66×10^{-1}
5	2.62×10^1	2.24×10^1	1.65×10^{-2}	1.65	7.04×10^{-1}
10	2.31×10^1	1.97×10^1	1.45×10^{-2}	1.45	7.18×10^{-1}
20	2.57×10^1	2.17×10^1	1.59×10^{-2}	1.59	6.51×10^{-1}
30	2.71×10^1	2.26×10^1	1.65×10^{-2}	1.65	6.12×10^{-1}
40	2.48×10^1	2.10×10^1	1.55×10^{-2}	1.54	6.73×10^{-1}
50	2.54×10^1	2.15×10^1	1.58×10^{-2}	1.58	6.59×10^{-1}

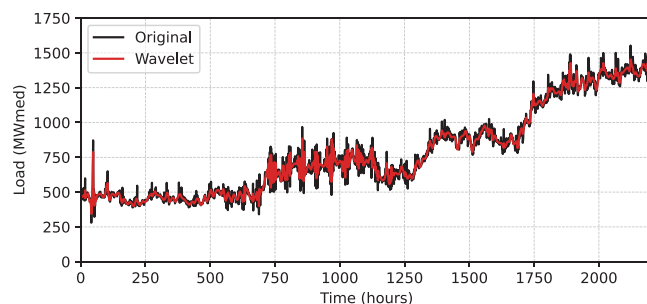


FIGURE 7 Wavelet transform versus observed values.

4.4 | Filtering evaluation

The wavelet transform must be applied without losing the signal's characteristics, which is why it is necessary to carefully evaluate the wavelet transform's response concerning the original signal. A high attenuation of the signal can result in a decomposed signal that does not represent the trend patterns of the original signal. An example of the result of the wavelet transform is shown in Figure 7.

The main parameter to be defined when applying the wavelet transform is the level at which it is applied. Levels higher than

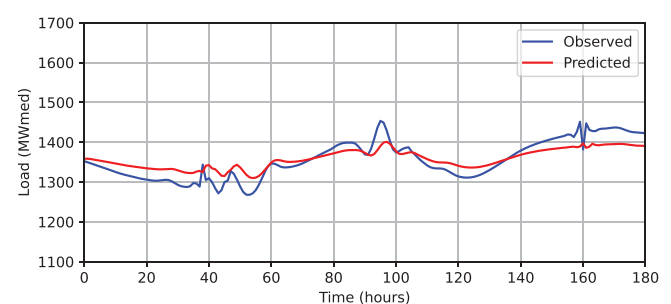


FIGURE 8 Comparative result of predicted by wavelet CNN-LSTM versus observed values.

50 did not result in a significant improvement in signal filtering, which is why they were not evaluated here. The results of applying the proposed wavelet CNN-LSTM model by varying the filtering level (L) are shown in Table 7.

Considering the signal used, a level greater than 10 did not result in an improvement in the decomposition, so this was the value set to be applied in this filter. The comparison of the original signal versus the predicted one using a level equal to 10 is presented in Figure 8. Based on the use of the model with its adjusted parameters, a comparative analysis of

TABLE 8 Benchmarking results.

Model	RMSE	MAE	MAPE	SMAPE	R ²	Time (s)
DeepNPTS	5.00×10^1	4.27×10^1	3.35×10^{-2}	3.31	1.47×10^{-2}	3.52
TCN	7.55×10^2	6.86×10^2	5.29×10^{-1}	4.04×10^1	2.30×10^2	3.91
NHITS	4.50×10^1	3.91×10^1	3.05×10^{-2}	3.03	1.79×10^{-1}	1.74
NBEATS	4.16×10^1	3.54×10^1	2.78×10^{-2}	2.75	2.98×10^{-1}	3.65
CNN-LSTM	4.79×10^1	3.51×10^1	2.58×10^{-2}	2.60	4.25×10^{-1}	3.26×10^2
Our	2.40×10^1	2.02×10^1	1.48×10^{-2}	1.48	6.96×10^{-1}	2.03×10^2

the proposed wavelet CNN-LSTM model in comparison with other structures is presented in the next subsection.

4.5 | Benchmarking results

The comparative analysis presented here aims to highlight the best structure for predicting the signal evaluated here. In addition to the proposed model, other structures are analysed considering the same dataset and division for training and testing the model. The results of this analysis are shown in Table 8.

Using a CNN-based model results in a longer time needed to train the model, but here, this is not a limiting factor considering that the forecasts are hourly, and the prediction result occurs in less than 5 min. The proposed wavelet CNN-LSTM model (our) had a superior result in all the evaluation error metrics, proving to be a promising alternative to be applied in biomass thermal systems electricity power forecasting.

5 | FINAL CONSIDERATIONS

The integration of wavelet, CNN, and LSTM approaches for time series forecasting of electricity power generation offers a promising avenue for enhancing predictive accuracy and efficiency. Through the fusion of convolutional and recurrent neural network architectures, the CNN-LSTM enables the extraction of spatial and temporal features from complex data streams, thereby capturing intricate patterns inherent in biomass-based power generation processes. By leveraging the inherent strengths of both CNNs and LSTMs, such as feature extraction and sequence modelling. This methodology facilitates robust forecasting capabilities crucial for optimizing energy generation and consumption strategies.

The utilization of biomass thermal systems underscores a sustainable pathway towards renewable energy production, aligning with broader environmental objectives. As research in this field continues to evolve, further advancements in model architectures, data preprocessing techniques, and domain-specific knowledge integration hold the potential to drive transformative innovations in renewable energy forecasting and management.

Given a MAPE of 0.0148, the wavelet CNN-LSTM showed promise for predicting electricity from biomass generation. The

variation in the amount of data used to train or test the model had a significant influence on the performance results, as did the use of other optimizers or learning rates. Considering the appropriate use of these parameters, the model had promising results in the evaluations presented here, proving to be a robust model even when several experiments are evaluated according to the statistical analysis presented here.

The use of the wavelet transform is an interesting solution as it reduces the complexity of the analysis since abrupt variations are not considered due to the filtering of the signal. Using a hybrid method that considers the filtered signal in the input is a promising approach, validating the proposed model presented here.

There are several avenues for future research and development in CNN-LSTM time series forecasting. Using other CNN models to detect signal characteristics could be an alternative to improve prediction. Considering that humidity and temperature have an influence on the level of dams in hydroelectric plants, and the use of thermal generation is related to the lack of available water for the hydroelectric, it would be promising to use this information as co-features to improve the predictions on each of the power plants.

AUTHOR CONTRIBUTIONS

William Gouvêa Buratto and Rafael Ninno Muniz wrote the introduction, method, results, and conclusion sections, drew the figures, and executed the experiments. Ademir Nied, Carlos Frederico de Oliveira Barros, Rodolfo Cardoso, and Gabriel Villarrubia Gonzalez proofread and supervised.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Available on request to the corresponding author.

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